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Optimization for Machine Learning — CS-439 - IC
18.06.2026 from 9h15 to 12h15
Duration : 180 minutes

Student One

SCIPER: 111111

Wait for the start of the exam before turning to the next page. This document is printed double sided, 24 pages. Do not unstaple.

- This is a closed book exam. No electronic devices of any kind.
- Place on your desk: your student ID, writing utensils, one double-sided A4 page cheat sheet if you have one; place all other personal items below your desk or on the side.
- You each have a different exam.
- For technical reasons, **do use black or blue pens for the MCQ part, no pencils!** Use white corrector if necessary.

Respectez les consignes suivantes Observe this guidelines Beachten Sie bitte die unten stehenden Richtlinien		
choisir une réponse select an answer Antwort auswählen	ne PAS choisir une réponse NOT select an answer NICHT Antwort auswählen	Corriger une réponse Correct an answer Antwort korrigieren
  		 
ce qu'il ne faut PAS faire what should NOT be done was man NICHT tun sollte		
     		



First part, multiple choice

There is **exactly one** correct answer per question.

Question 1 (Convexity) Let $X \subseteq \mathbb{R}^d$ be an open, convex set, and let $f : X \rightarrow \mathbb{R}$ be a continuously differentiable function. Consider the following statements.

- (A) Every sublevel set $\{\mathbf{x} \in X : f(\mathbf{x}) \leq \alpha\}$ is convex. That is, every set that contains all points $\mathbf{x}$ where the function value is at most α is convex.
- (B) The epigraph $\{(\mathbf{x}, t) \in X \times \mathbb{R} : t \geq f(\mathbf{x})\}$ is convex. That is, the region that contains all points lying on or above the graph of f is convex.
- (C) For every coordinate vector e_i , every $\mathbf{x} \in X$, and every $t \in \mathbb{R}$ such that $\mathbf{x} + te_i \in X$, we have

$$(\nabla f(\mathbf{x} + te_i) - \nabla f(\mathbf{x}))^\top (te_i) \geq 0.$$

Which statement is equivalent to the convexity of f ?

- ☐ None of (A), (B), or (C) is equivalent to convexity of f .
- ☐ (A), (B), and (C) are all equivalent to convexity of f .
- ☐ Only (A) is equivalent to convexity of f .
- ☐ (A) and (C) are equivalent to convexity of f , but (B) is not.
- ☐ Only (B) is equivalent to convexity of f .
- ☐ (B) and (C) are equivalent to convexity of f , but (A) is not.
- ☐ Only (C) is equivalent to convexity of f .
- ☐ (A) and (B) are equivalent to convexity of f , but (C) is not.



Question 2 (Gradient Descent) Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable and L -smooth. Assume f is bounded below, i.e., $\exists m \in \mathbb{R}$ such that $f(x) \geq m$, $\forall x \in \mathbb{R}^d$. Run gradient descent with $\gamma = 1/L$:

$$x_{t+1} = x_t - \frac{1}{L} \nabla f(x_t).$$

Consider the following statements:

(A) $\sum_{t=0}^{\infty} \|x_{t+1} - x_t\|^2 < \infty$.

(B) For every $T \geq 1$,

$$\min_{0 \leq t < T} \|\nabla f(x_t)\|^2 \leq \frac{2L(f(x_0) - f_{\inf})}{T},$$

where $f_{\inf}$ is any lower bound of f .

(C) For every subsequence $\{x_{t_k}\}_{k=1}^{\infty}$ with $t_k \rightarrow \infty$, if there exists $\bar{x} \in \mathbb{R}^d$ such that $\lim_{k \rightarrow \infty} x_{t_k} = \bar{x}$, then $\bar{x}$ is a local minimum of f .

Which choice correctly identifies all statements that are guaranteed to hold?

Hint: recall the sufficient-decrease (descent) lemma for L -smooth functions, $f(x_{t+1}) \leq f(x_t) - \frac{1}{2L} \|\nabla f(x_t)\|^2$.

- ☐ Only (A) and (B) are guaranteed.
- ☐ All three statements are guaranteed.
- ☐ Only (B) is guaranteed.
- ☐ Only (B) and (C) are guaranteed.
- ☐ Only (C) is guaranteed.
- ☐ None of the statements is guaranteed.
- ☐ Only (A) is guaranteed.
- ☐ Only (A) and (C) are guaranteed.



Question 3 (Projected Gradient Descent) Let $X \subseteq \mathbb{R}^d$ be nonempty, closed, and convex, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable. Let

$$x^+ = \Pi_X(x - \gamma \nabla f(x)), \quad \gamma > 0,$$

where $x \in X$. Define the actual PGD displacement

$$d = x^+ - x.$$

Consider the following statements:

(A) $\nabla f(x)^\top d \leq -\frac{1}{2\gamma} \|d\|^2.$

(B) If $x^+ \in \partial X$ and f is convex and L -smooth, and $0 < \gamma \leq 1/L$, then x^+ is a minimizer of f over X .

(C) Let

$$x_{\text{GD}}^+ = x - \gamma \nabla f(x), \quad x_{\text{PGD}}^+ = \Pi_X(x - \gamma \nabla f(x)).$$

If f is convex and L -smooth, and $0 < \gamma \leq 1/L$, then

$$f(x_{\text{GD}}^+) \leq f(x_{\text{PGD}}^+).$$

Which choice correctly identifies all statements that are guaranteed to hold?

- ☐ Only (A) is guaranteed.
- ☐ None of the statements is guaranteed.
- ☐ Only (B) is guaranteed.
- ☐ Both (A) and (B) are guaranteed.
- ☐ All three statements are guaranteed.
- ☐ Both (B) and (C) are guaranteed.
- ☐ Only (C) is guaranteed.
- ☐ Both (A) and (C) are guaranteed.

**Question 4** (Proximal Gradient Descent) Let

$$f(x) = g(x) + h(x), \quad g(x) = \frac{1}{2}\|x - a\|^2, \quad h(x) = \lambda\|x\|_1,$$

where $\lambda > 0$ and $x, a \in \mathbb{R}^d$. We denote $x^{(t)}$ as the t -th iterate, and $x_i^{(t)}$ as its i -th coordinate. Proximal gradient descent is run with stepsize $\gamma = 1$:

$$x^{(t+1)} = \text{prox}_{\gamma h}(x^{(t)} - \gamma \nabla g(x^{(t)})).$$

For a target accuracy $\varepsilon > 0$, say that an iterate $x^{(t)}$ is ε -optimal if

$$f(x^{(t)}) - f(x^*) \leq \varepsilon,$$

where x^* is a global minimizer of f . Which statement is correct?

- ☐ The first iterate $x^{(1)}$ is the global minimizer, independently of $x^{(0)}$, and its coordinates are $x_i^{(1)} = x_i^* = \text{sgn}(a_i) \max\{|a_i| - \lambda, 0\}$.
- ☐ The proximal update uses hard-thresholding: $x_i^{(t+1)} = a_i \mathbf{1}_{\{|a_i| > \lambda\}}$.
- ☐ The update is ordinary gradient descent on $g + h$, so it is undefined whenever some coordinate $x_i^{(t)}$ is zero.
- ☐ For sufficiently small $\varepsilon > 0$, proximal gradient descent requires more than one iteration to obtain an ε -optimal point.
- ☐ The first iterate is $x^{(1)} = a$, independently of $x^{(0)}$.

Question 5 (Subgradients) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be

$$f(x_1, x_2) = \max\{x_1, x_2, 0\}.$$

At $(0, 0)$, which of the following sets contains the largest number of vectors belonging to $\partial f(0, 0)$ while ensuring all of the elements belong to $\partial f(0, 0)$?

- ☐ $\{(0, 0), (0.3, 0.3), (0.2, 0.7)\}$.
- ☐ $\{(0.3, 0.3), (0.6, -0.6), (0.5, 0.5), (0.1, 0.4)\}$.
- ☐ $\{(0.2, 0.7), (0.4, 0.4), (0.8, 0.3), (0.6, -0.1)\}$.
- ☐ $\{(0.3, 0.3), (0.6, 0.6), (1, 0), (0.2, 0.9)\}$.
- ☐ $\{(0, 0), (0.4, 0.4), (0.6, 0.2), (1, 1)\}$.
- ☐ No nonzero vector belongs to $\partial f(0, 0)$ because f is not differentiable at $(0, 0)$.



Question 6 (Gradient Descent) Consider the problem of estimating an unknown parameter vector $\theta \in \mathbb{R}^d$ from a dataset $\mathcal{D}$.

Suppose that the likelihood of the dataset under parameters θ is given by a known function

$$q(\mathcal{D} \mid \theta) = f(\theta).$$

where $f > 0$ and $-\log f$ is convex and differentiable.

In addition, assume that each coordinate has an independent Laplace prior centered at a known value μ_i :

$$\theta_i \sim \text{Laplace}(\mu_i, b),$$

whose density is

$$p(\theta_i) = \frac{1}{2b} \exp\left(-\frac{|\theta_i - \mu_i|}{b}\right).$$

The maximum a posteriori (MAP) estimator is defined as

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta \in \mathbb{R}^d} q(\theta \mid \mathcal{D}).$$

Which of the following optimization problems correctly formulates the MAP estimator for θ ?

- ☐ $\min_{\theta \in \mathbb{R}^d} \left[-\log f(\theta) + \frac{1}{b} \|\theta - \mu\|_1 \right].$
- ☐ $\min_{\theta \in \mathbb{R}^d} \left[-\log f(\theta) + \frac{1}{b} \|\theta - \mu\|_2^2 \right].$
- ☐ $\min_{\theta \in \mathbb{R}^d} \left[-\log f(\theta) + b \|\theta - \mu\|_2^2 \right].$
- ☐ $\min_{\theta \in \mathbb{R}^d} \left[-f(\theta) + \frac{1}{b} \|\theta\|_1 \right].$
- ☐ $\max_{\theta \in \mathbb{R}^d} \left[f(\theta) - \frac{1}{b} \|\theta - \mu\|_1 \right].$
- ☐ $\max_{\theta \in \mathbb{R}^d} \left[\log f(\theta) + \frac{1}{b} \|\theta - \mu\|_1 \right].$

Question 7 (Subgradients) Consider the MAP optimization objective F that you obtained in the previous question. Which of the following conditions is necessary and sufficient for μ to be a minimizer of F ?

- ☐ $\|\nabla f(\mu)\|_\infty \leq \frac{1}{b}.$
- ☐ $\|\nabla \log f(\mu)\|_2^2 \leq \frac{1}{b}.$
- ☐ $\|\nabla \log f(\mu)\|_\infty \geq \frac{1}{b}.$
- ☐ $\nabla \log f(\mu) = 0.$
- ☐ $\|\nabla \log f(\mu)\|_\infty \leq \frac{1}{b}.$
- ☐ $\|\nabla \log f(\mu)\|_1 \leq \frac{1}{b}.$



Question 8 (Gradient Descent) Consider the function

$$f(x, y) = \frac{1}{m}(x + y)^2,$$

where $m > 0$. We run gradient descent with stepsize

$$\eta = \frac{m}{4}.$$

Starting from an arbitrary point $(x_0, y_0) \in \mathbb{R}^2$, how many gradient descent steps are needed in the worst case to reach a point (x_t, y_t) such that $f(x_t, y_t) - f(x^*, y^*) < \varepsilon$?

- ☐ 1
- ☐ m
- ☐ $\Theta(\sqrt{1/\varepsilon})$
- ☐ $\Theta(\log(1/\varepsilon))$
- ☐ All other options are incorrect.
- ☐ $\Theta(m \log(1/\varepsilon))$

Question 9 (Coordinate Descent) Consider the function

$$f(x, y) = x^2 + 2y^2 + xy.$$

We want to minimize f using a coordinate descent method that updates one coordinate at a time. We are allowed to pick a coordinate-wise stepsize η_i as we like.

Which rule for choosing coordinates among the ones listed below gives the best standard worst-case convergence guarantee?

- ☐ Choose both coordinates uniformly:
$$p_x = p_y = \frac{1}{2}.$$
- ☐ Use deterministic coordinate descent: update x and y consecutively in a loop.
- ☐ No convergence guarantee can be obtained for coordinate descent on this function.
- ☐ Sample coordinates with probabilities:

$$p_x = \frac{2}{3}, \quad p_y = \frac{1}{3}.$$

- ☐ Only update y .
- ☐ Sample coordinates with probabilities:

$$p_x = \frac{1}{3}, \quad p_y = \frac{2}{3}.$$



Question 10 (Frank-Wolfe / LMO) Let $F : \mathbb{R}^d \rightarrow \mathbb{R}$ be a loss function and let $\|\cdot\|$ denote any norm on $\mathbb{R}^d$. Define the dual norm and the Linear Minimization Oracle by

$$\|v\|_* = \max_{\|u\| \leq 1} \langle u, v \rangle, \quad \text{LMO}(v) = \operatorname{argmin}_{\|u\| \leq 1} \langle u, v \rangle.$$

Let g_t denote the gradient estimate at step t ; assume $g_t \neq 0$. Let u_t denote any element of $\text{LMO}(g_t)$, and let $\eta_1, \eta_2 > 0$. Consider two updates built from the first-order Taylor model of F around w_t :

Algorithm 1:

$$w_{t+1} = \operatorname{argmin}_{\substack{w \in \mathbb{R}^d \\ \|w - w_t\| \leq \eta_1}} \{F(w_t) + \langle g_t, w - w_t \rangle\}$$

Algorithm 2:

$$w_{t+1} = \operatorname{argmin}_{w \in \mathbb{R}^d} \left\{ F(w_t) + \langle g_t, w - w_t \rangle + \frac{1}{2\eta_2} \|w - w_t\|^2 \right\}$$

Which statement about the two updates is correct?

- ☐ Algorithm 1 produces a better update because its iterate is constrained to a ball.
- ☐ Both updates move along an LMO direction $u_t \in \text{LMO}(g_t)$ and differ only in their step sizes.
- ☐ Algorithm 1 steps in the direction $-g_t$ (the negative gradient, rescaled to the ball boundary).
- ☐ Algorithm 2 produces a better update because it incorporates a second-order Taylor approximation of F .

Question 11 (Frank-Wolfe / LMO) Using the notation from the previous question, the closed-form update of Algorithm 2 is:

- ☐ The update cannot be written via the LMO; it depends on $\mathbb{E}[g_t]$.
- ☐ $w_{t+1} = w_t + \eta_2 u_t$, for $u_t \in \text{LMO}(g_t)$.
- ☐ $w_{t+1} = w_t + \eta_2 \|g_t\|_* u_t$, for $u_t \in \text{LMO}(g_t)$.
- ☐ $w_{t+1} = w_t + \frac{1}{2\eta_2} \|g_t\|_* u_t$, for $u_t \in \text{LMO}(g_t)$.

Question 12 (Frank-Wolfe / LMO) Define two oracles for the same norm $\|\cdot\|$, differing only in their feasible set:

$$\text{LMO}_1(v) = \operatorname{argmin}_{\|u\|=1} \langle u, v \rangle \quad (\text{unit sphere}), \quad \text{LMO}_2(v) = \operatorname{argmin}_{\|u\| \leq 1} \langle u, v \rangle \quad (\text{closed unit ball}).$$

Which statement is correct?

- ☐ LMO_2 always returns $u = 0$, since $u = 0$ lies inside the ball and minimizes any norm.
- ☐ The two oracles agree only when $\|\cdot\|$ is the Euclidean norm $\|\cdot\|_2$.
- ☐ LMO_1 and LMO_2 always return different solutions: the ball allows interior points that cannot be returned by LMO_1 .
- ☐ For every $v \neq 0$, any minimizer of $\text{LMO}_2(v)$ lies on the unit sphere and is therefore also a valid minimizer of $\text{LMO}_1(v)$; the two oracles are equivalent.



Question 13 (Affine Invariance) Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice differentiable with invertible Hessian at every iterate. Let $\mathbf{A} \in \mathbb{R}^{d \times d}$ be invertible, define $h(\mathbf{y}) = f(\mathbf{A}\mathbf{y})$, and set $\mathbf{y}_0 = \mathbf{A}^{-1}\mathbf{x}_0$. Run gradient descent (with step size $\eta > 0$) and Newton's method each on $f(\mathbf{x})$ and on $h(\mathbf{y})$ from the respective starting points. For which algorithm(s) do the iterates satisfy $\mathbf{x}_t = \mathbf{A}\mathbf{y}_t$ for all $t \geq 0$ for any invertible $\mathbf{A}$?

- ☐ Only Newton's method.
- ☐ Both gradient descent and Newton's method.
- ☐ Neither gradient descent nor Newton's method.
- ☐ Only gradient descent.

Question 14 (Smoothness) Recall that $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth over a convex set X if

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{x} - \mathbf{y}\|^2 \quad \forall \mathbf{x}, \mathbf{y} \in X.$$

A student attempts to prove: if f is twice continuously differentiable and L -smooth over an open convex X , then $\lambda_{\max}(\nabla^2 f(\mathbf{x})) \leq L$ for all $\mathbf{x} \in X$.

- S1.** Since f is L -smooth, ∇f is L -Lipschitz: $\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in X$.
- S2.** Fix $\mathbf{x} \in X$ and unit $\mathbf{v}$. Since X is open, $\mathbf{x} + t\mathbf{v} \in X$ for small $t > 0$, hence $\|\nabla f(\mathbf{x} + t\mathbf{v}) - \nabla f(\mathbf{x})\| \leq Lt$.
- S3.** Dividing by $t > 0$ and letting $t \rightarrow 0^+$:

$$\|\nabla^2 f(\mathbf{x})\mathbf{v}\| = \lim_{t \rightarrow 0^+} \frac{\|\nabla f(\mathbf{x} + t\mathbf{v}) - \nabla f(\mathbf{x})\|}{t} \leq L.$$

- S4.** Since this holds for every unit $\mathbf{v}$, $\|\nabla^2 f(\mathbf{x})\|_{\text{op}} \leq L$, hence $\lambda_{\max}(\nabla^2 f(\mathbf{x})) \leq \|\nabla^2 f(\mathbf{x})\|_{\text{op}} \leq L$. □

Where is the error, if any?

- ☐ **S3:** the limit and the norm are exchanged without justification; since $\|\cdot\|$ is not linear, this interchange is not valid in general.
- ☐ **S4:** the inequality $\lambda_{\max}(H) \leq \|H\|_{\text{op}}$ holds only when $\nabla^2 f(\mathbf{x})$ is positive semidefinite, which is not assumed.
- ☐ There is no error; the proof is correct as written.
- ☐ **S1:** the implication goes the wrong way. L -Lipschitz gradient implies L -smoothness, but L -smoothness alone does not imply L -Lipschitz gradient for a non-convex f .



Question 15 (Smoothness-convexity) Recall that $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth over a convex set X if

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{x} - \mathbf{y}\|^2, \quad \forall \mathbf{x}, \mathbf{y} \in X.$$

Consider the following statements.

- (A) Every convex function is smooth.
- (B) Every smooth function is convex.
- (C) Every non-convex differentiable function is smooth.
- (D) Every differentiable concave function is smooth with $L = 0$.
- (E) If f is L -smooth over X , then it is also L' -smooth over X for every $L' \geq L$.
- (F) Every twice-differentiable function satisfying $\lambda_{\max}(\nabla^2 f(\mathbf{x})) \leq L$ for all $\mathbf{x} \in X$ is L -smooth over X .

Which choice correctly identifies all true statements?

- ☐ Only (D) and (E) are true.
- ☐ Only (A) and (F) are true.
- ☐ All six statements are true.
- ☐ Only (D), (E), and (F) are true.
- ☐ Only (E) and (F) are true.
- ☐ Only (B), (D), and (F) are true.
- ☐ (A), (D), (E), and (F) are true.
- ☐ Only (E) is true.

Question 16 (SGD) Consider $f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x})$, μ -strongly convex with unique global minimizer $\mathbf{x}^*$. SGD samples $i \in [n]$ uniformly at random and updates $\mathbf{x}_{t+1} := \mathbf{x}_t - \gamma_t \mathbf{g}_t$ where $\mathbf{g}_t = \nabla f_i(\mathbf{x}_t)$. Assume $\mathbb{E}[\|\mathbf{g}_t\|^2] \leq B^2$ for all t . Suppose the *interpolation condition* holds:

$$\nabla f_i(\mathbf{x}^*) = \mathbf{0} \quad \text{for every } i \in [n].$$

What is the tightest upper bound on $\mathbb{E}[\|\mathbf{g}_t\|^2 \mid \mathbf{x}_t = \mathbf{x}^*]$?

- ☐ B^2/μ
- ☐ B^2
- ☐ B^2/n
- ☐ 0



Question 17 (SGD) Continuing with the same setup as the previous question, assume $B > 0$. The following inequality can be shown by combining the update rule with the strong convexity of f ; you may use it without proof: for any **constant** step-size $\gamma \in (0, 1/\mu)$,

$$\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2] \leq (1 - \mu\gamma) \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}^*\|^2] + \gamma^2 \mathbb{E}[\|\mathbf{g}_t\|^2].$$

You may also use: for any $r \in (0, 1)$, $\sum_{s=0}^{T-1} r^s \leq \frac{1}{1-r}$.

Applying this inequality repeatedly with a constant step-size $\gamma \in (0, 1/\mu)$, what can be definitively concluded from the derived bound about $\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2]$ as $T \rightarrow \infty$?

- ☐ The inequality diverges to infinity unless a decreasing step-size $\gamma_t = \mathcal{O}(1/t)$ is used.
- ☐ The sequence is guaranteed to converge to 0 exponentially fast, because the interpolation condition ensures $\mathbb{E}[\|\mathbf{g}_t\|^2] = 0$ for all t .
- ☐ The upper bound decays to 0 at a rate of $\mathcal{O}(1/T)$, matching the standard strongly convex SGD rate.
- ☐ The upper bound converges to $\gamma B^2/\mu > 0$. Therefore, the bound only guarantees convergence to a neighborhood, not to zero.

Question 18 (SGD) Continuing with the same setup, now additionally assume each f_i is convex and L -smooth. You may use the following without proof:

- **Co-coercivity:** for any convex L -smooth function h and any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$\|\nabla h(\mathbf{x}) - \nabla h(\mathbf{y})\|^2 \leq L (\nabla h(\mathbf{x}) - \nabla h(\mathbf{y}))^\top (\mathbf{x} - \mathbf{y}).$$

- **Strong convexity of f :** $\nabla f(\mathbf{x})^\top (\mathbf{x} - \mathbf{x}^*) \geq \mu \|\mathbf{x} - \mathbf{x}^*\|^2$.
- **Expansion (from the update rule + unbiasedness):** $\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2] = \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}^*\|^2] - 2\gamma \mathbb{E}[\nabla f(\mathbf{x}_t)^\top (\mathbf{x}_t - \mathbf{x}^*)] + \gamma^2 \mathbb{E}[\|\mathbf{g}_t\|^2]$.

Under the interpolation condition with **constant** step-size $\gamma \in (0, 1/L]$, applying these inequalities, what can be definitively concluded about $\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2]$?

- ☐ The upper bound decays to a positive noise floor $\frac{\gamma B^2}{2}$ exponentially fast:
$$\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \leq \left(1 - \mu\gamma\left(1 - \frac{\gamma L}{2}\right)\right)^T \|\mathbf{x}_0 - \mathbf{x}^*\|^2 + \frac{\gamma B^2}{2}.$$
- ☐ The upper bound decays to 0 exponentially fast:
$$\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \leq \left(1 - 2\mu\gamma\left(1 - \frac{\gamma L}{2}\right)\right)^T \|\mathbf{x}_0 - \mathbf{x}^*\|^2.$$
- ☐ The upper bound decays to 0 at a polynomial rate:
$$\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \leq \frac{1}{1+2\mu\gamma LT} \|\mathbf{x}_0 - \mathbf{x}^*\|^2.$$
- ☐ The upper bound decays to a positive noise floor $\frac{\gamma B^2}{\mu}$ at a polynomial rate:
$$\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \leq \frac{1}{1+\mu\gamma T} \|\mathbf{x}_0 - \mathbf{x}^*\|^2 + \frac{\gamma B^2}{\mu}.$$



Second part, true/false questions

Question 19 (Projected Gradient Descent) Let $X \subseteq \mathbb{R}^d$ be a closed set. Define

$$\Pi_X(z) \in \operatorname{argmin}_{y \in X} \|y - z\|.$$

Since $\Pi_X(z)$ is the closest feasible point to z , the projection map is nonexpansive:

$$\|\Pi_X(u) - \Pi_X(v)\| \leq \|u - v\| \quad \forall u, v \in \mathbb{R}^d.$$

☐ TRUE ☐ FALSE

Question 20 (Gradient Descent) Subgradient descent $\mathbf{x}_{t+1} = \mathbf{x}_t - \gamma \mathbf{g}_t$ (with $\mathbf{g}_t \in \partial f(\mathbf{x}_t)$) on a convex function is a descent method: it satisfies $f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t)$ at every iteration.

☐ TRUE ☐ FALSE

Question 21 (Subgradients) If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable at $\mathbf{x}$, then $\partial f(\mathbf{x}) = \{\nabla f(\mathbf{x})\}$.

☐ TRUE ☐ FALSE

Question 22 (Subgradients) For a convex function f , every subgradient $\mathbf{g} \in \partial f(\mathbf{x})$ can be written as a limit $\mathbf{g} = \lim_{n \rightarrow \infty} \nabla f(\mathbf{x}_n)$ of gradients at differentiable points $\mathbf{x}_n \rightarrow \mathbf{x}$.

☐ TRUE ☐ FALSE

Question 23 (Proximal Gradient Descent) Let $h(x) = \lambda \|x\|_1$, where $\lambda > 0$. For a stepsize $\gamma > 0$, the proximal operator is

$$\operatorname{prox}_{\gamma h}(y) = \arg \min_{u \in \mathbb{R}^d} \left\{ \frac{1}{2\gamma} \|u - y\|^2 + h(u) \right\}.$$

Wherever h is differentiable at y , the proximal update

$$u^+ = \operatorname{prox}_{\gamma h}(y)$$

is equivalent to one ordinary gradient descent step on h :

$$u^+ = y - \gamma \nabla h(y).$$

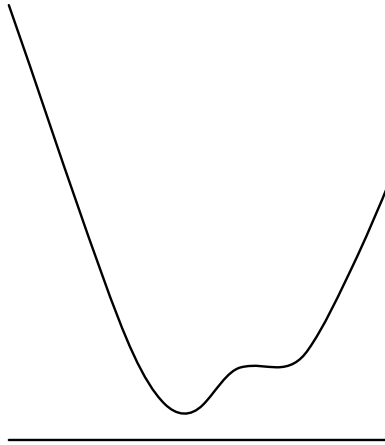
☐ TRUE ☐ FALSE

Question 24 (Convexity) If a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex, then for every $\mathbf{x}, \mathbf{v} \in \mathbb{R}^d$, the function $g : t \mapsto f(\mathbf{x} + t\mathbf{v})$ is convex. The reverse direction is also correct: if for every $\mathbf{x}, \mathbf{v} \in \mathbb{R}^d$, the function $g_{\mathbf{x}, \mathbf{v}} : t \mapsto f(\mathbf{x} + t\mathbf{v})$ is convex, then f is convex.

☐ TRUE ☐ FALSE



Question 25

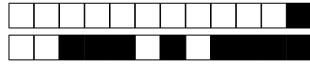


Does the function f in the image above satisfy the Polyak-Łojasiewicz inequality?

☐ TRUE

☐ FALSE

DRAFT



Third part, open questions

Answer in the space provided! Your answer must be justified with all steps. Do not cross any checkboxes, they are reserved for correction.

Question 26: 2 points.

Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$. Its dual norm is defined as

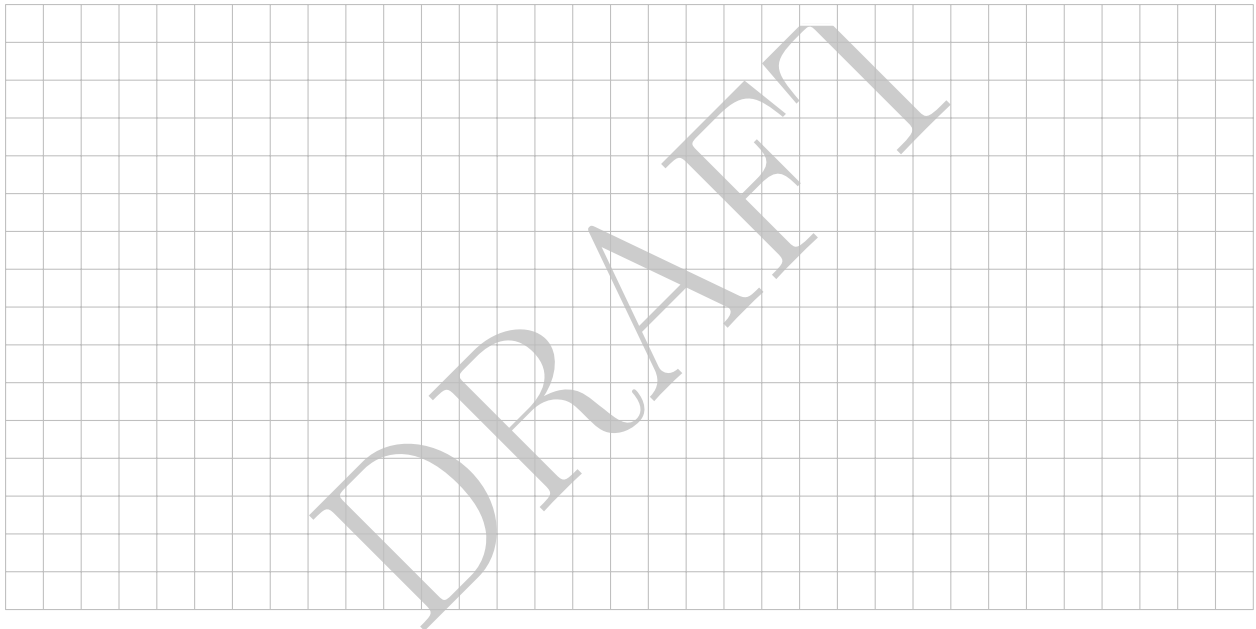
$$\|\mathbf{x}\|_* = \max_{\|\mathbf{s}\| \leq 1} \langle \mathbf{x}, \mathbf{s} \rangle.$$

You may use the fact that the dual norm operation is an involution, i.e., $\|\mathbf{x}\|_{**} = \|\mathbf{x}\|$.

Prove that for every $\mathbf{x} \in \mathbb{R}^d$,

$$\|\mathbf{x}\| = \max_{\|\mathbf{s}\|_* = 1} \langle \mathbf{s}, \mathbf{x} \rangle.$$

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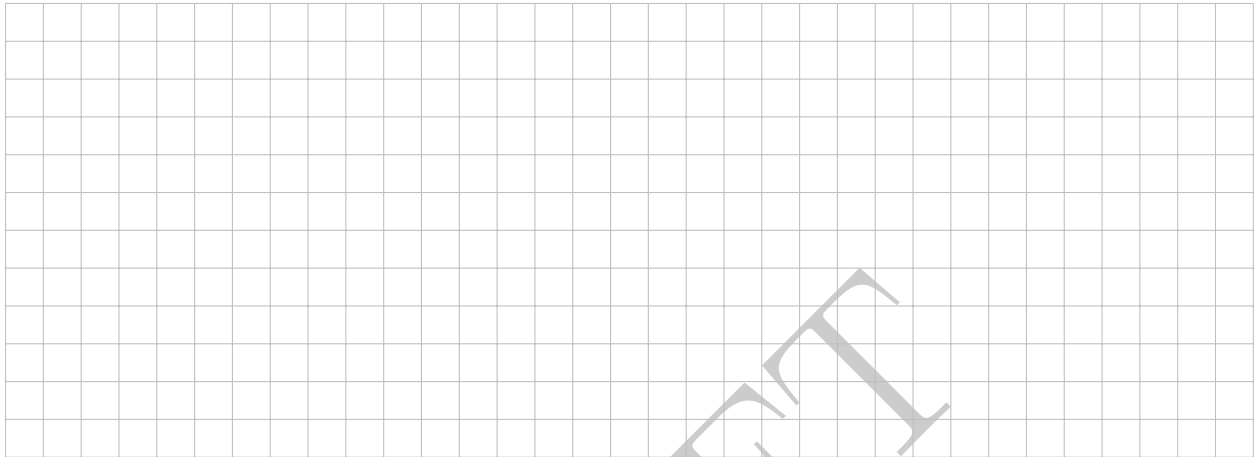
Question 27: 2 points.

Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$, and let $\|\cdot\|_*$ be its dual norm. Fix $\mathbf{x} \neq 0$ and define

$$A(\mathbf{x}) = \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1, \langle \mathbf{s}, \mathbf{x} \rangle = \|\mathbf{x}\|\}.$$

Prove that $A(\mathbf{x})$ is convex.

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**Question 28:** 2 points.

Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$, and let $\|\cdot\|_*$ be its dual norm. From the previous questions, you may use the identity

$$\|\mathbf{x}\| = \max_{\|\mathbf{s}\|_* = 1} \langle \mathbf{s}, \mathbf{x} \rangle,$$

and the fact that for every $\mathbf{x} \neq 0$, the set

$$\{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1, \langle \mathbf{s}, \mathbf{x} \rangle = \|\mathbf{x}\|\}$$

is convex.

You may also use Danskin's theorem: if

$$f(\mathbf{x}) = \max_{i \in I} f_i(\mathbf{x}),$$

where I is compact and each f_i is convex and differentiable, then

$$\partial f(\mathbf{x}) = \text{conv} \left(\bigcup_{i \in I(\mathbf{x})} \partial f_i(\mathbf{x}) \right),$$

where

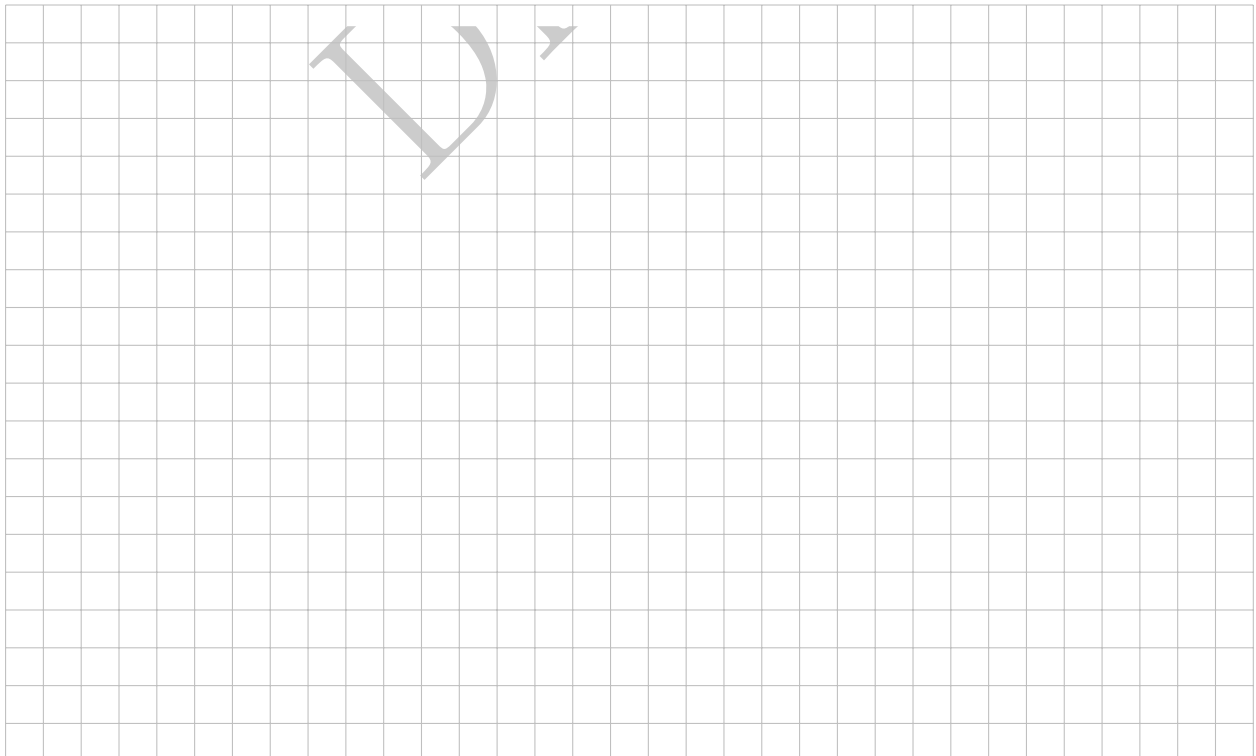
$$I(\mathbf{x}) = \{i \in I : f_i(\mathbf{x}) = f(\mathbf{x})\}$$

is the set of active maximizers.

Use Danskin's theorem to prove the general form for the subgradient of a norm:

$$\partial \|\mathbf{x}\| = \begin{cases} \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* \leq 1\}, & \text{if } \mathbf{x} = 0, \\ \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1, \langle \mathbf{s}, \mathbf{x} \rangle = \|\mathbf{x}\|\}, & \text{if } \mathbf{x} \neq 0. \end{cases}$$

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**Question 29:** 2 points.

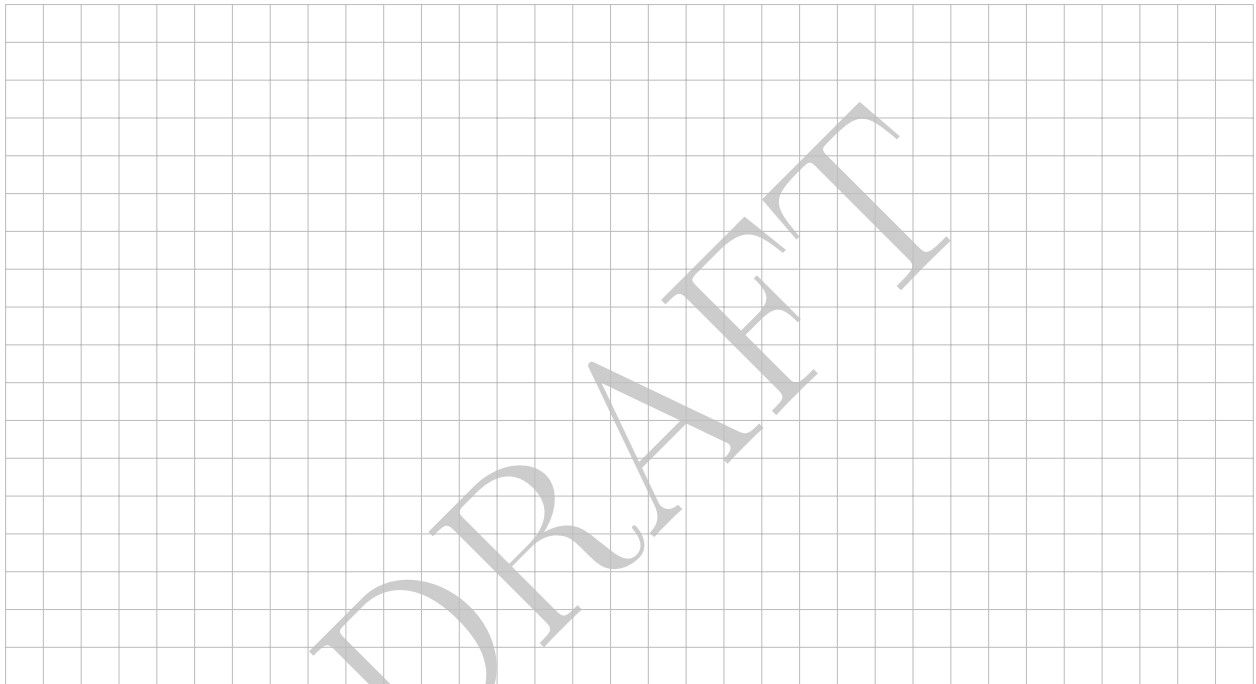
Let $H \in \mathbb{R}^{d \times d}$ be symmetric positive definite, and suppose all of its eigenvalues lie in $[\mu, L]$ with $0 < \mu \leq L$. Prove the *Kantorovich inequality*: for every unit vector $z \in \mathbb{R}^d$ (i.e. $\|z\|_2 = 1$),

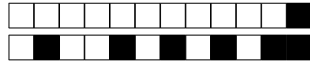
$$(z^\top H z)(z^\top H^{-1} z) \leq \frac{(L + \mu)^2}{4\mu L},$$

and show that this bound is attained, so that

$$\sup_{\|z\|_2=1} (z^\top H z)(z^\top H^{-1} z) = \frac{(L + \mu)^2}{4\mu L}.$$

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**Question 30:** 2 points.

Consider the least-squares objective

$$F(\theta) = \frac{1}{2n} \|\Phi\theta - y\|_2^2, \quad \Phi \in \mathbb{R}^{n \times d}, \quad y \in \mathbb{R}^n,$$

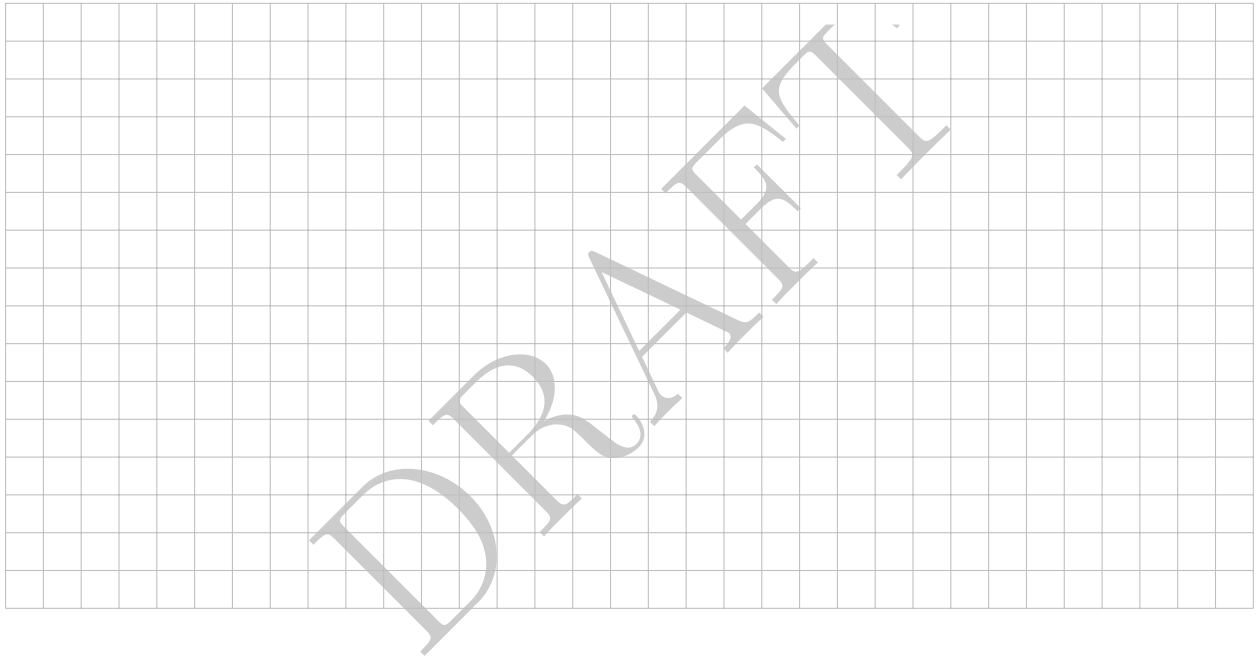
whose gradient is $\nabla F(\theta) = \frac{1}{n} \Phi^\top (\Phi\theta - y) = H(\theta - \eta_*)$, where $H = \frac{1}{n} \Phi^\top \Phi \in \mathbb{R}^{d \times d}$ is the (constant) Hessian of F and η_* is any minimizer of F . Starting from θ_{t-1} , run one step of gradient descent with step size γ_t ,

$$\theta_t = \theta_{t-1} - \gamma_t \nabla F(\theta_{t-1}).$$

Assume $\nabla F(\theta_{t-1}) \neq 0$. Show that the step size γ_t that minimizes $F(\theta_t)$ (exact line search) is

$$\gamma_t = \frac{\|\nabla F(\theta_{t-1})\|_2^2}{\nabla F(\theta_{t-1})^\top H \nabla F(\theta_{t-1})}.$$

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**Question 31:** 3 points.

Consider the same least-squares objective $F(\theta) = \frac{1}{2n} \|\Phi\theta - y\|_2^2$ as in the previous question, with constant Hessian $H = \frac{1}{n} \Phi^\top \Phi$ and minimizer η_* , and run gradient descent $\theta_t = \theta_{t-1} - \gamma_t \nabla F(\theta_{t-1})$ with the exact-line-search step size $\gamma_t = \|\nabla F(\theta_{t-1})\|_2^2 / (\nabla F(\theta_{t-1})^\top H \nabla F(\theta_{t-1}))$ from the previous question. Assume $\nabla F(\theta_{t-1}) \neq 0$. Assume in addition that F is strongly convex, i.e. H is invertible; denote by $0 < \mu \leq L$ the smallest and largest eigenvalues of H and by $\kappa = L/\mu \geq 1$ the condition number.

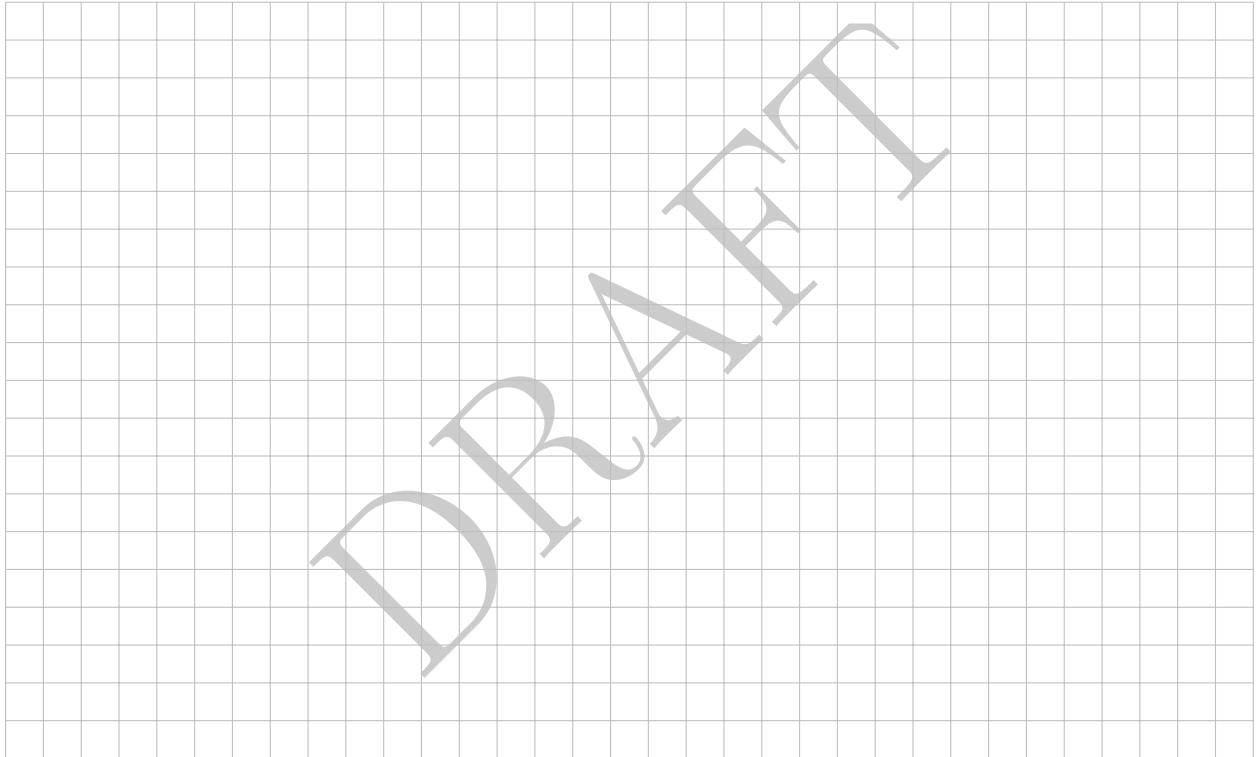
Show that

$$F(\theta_t) - F(\eta_*) \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^2 (F(\theta_{t-1}) - F(\eta_*)),$$

and compare this rate with that of gradient descent with a constant step size.

You may use the Kantorovich inequality established earlier: $\sup_{\|z\|_2=1} (z^\top H z)(z^\top H^{-1} z) = \frac{(L + \mu)^2}{4\mu L}$.

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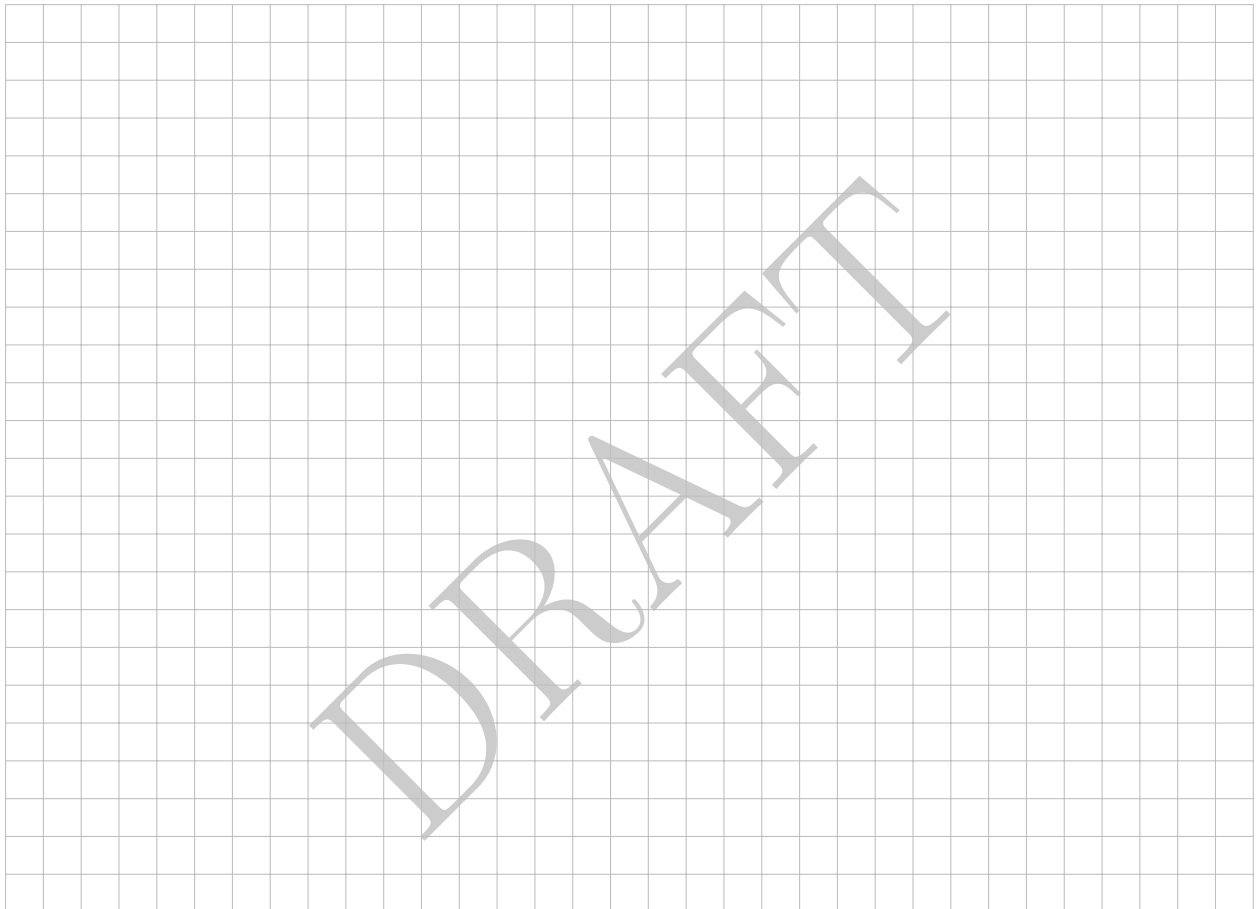
Question 32: 1 points.

In the previous two questions you analyzed two ways of choosing the step size for gradient descent on the least-squares objective F : the exact line search step size $\gamma_t = \arg \min_{\gamma \geq 0} F(\theta_{t-1} - \gamma \nabla F(\theta_{t-1}))$, and a constant step size (for instance $\gamma = 1/L$, or the optimal constant choice $\gamma = 2/(L + \mu)$).

Compare the two methods from a practical point of view. In particular, address:

- what each method needs to know about the problem (e.g. the constants μ and L); and
- what extra computation, if any, exact line search requires at each iteration.

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**Question 33:** 4 points.

Let $F : \mathbb{R}^d \rightarrow \mathbb{R}$ be μ -strongly convex and L -smooth, with unique minimizer η_* and condition number $\kappa = L/\mu$. For $\theta \neq \eta_*$, let α be the angle between the gradient $\nabla F(\theta)$ and the deviation to the optimum $\theta - \eta_*$, defined by

$$\cos \alpha = \frac{\nabla F(\theta)^\top (\theta - \eta_*)}{\|\nabla F(\theta)\|_2 \|\theta - \eta_*\|_2}$$

(equivalently, α is the angle between the descent direction $-\nabla F(\theta)$ and the direction $\eta_* - \theta$ pointing to the optimum).

You may use the following two facts directly, without proving them.

- *Kantorovich inequality. (established earlier)* If $H \in \mathbb{R}^{d \times d}$ is symmetric positive definite with all its eigenvalues in $[\mu, L]$, then for every $z \neq 0$,

$$(z^\top H z)(z^\top H^{-1} z) \leq \frac{(L + \mu)^2}{4\mu L} \|z\|_2^4.$$

- *Co-coercivity.* If $G : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex and β -smooth, then for all $\theta, \eta \in \mathbb{R}^d$,

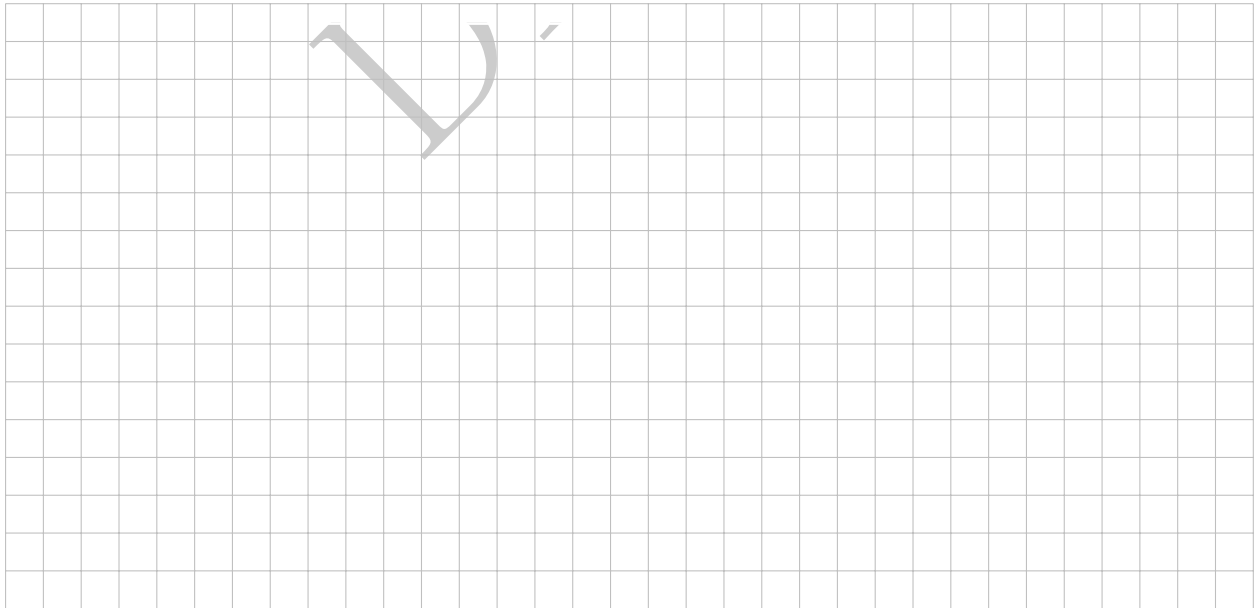
$$\frac{1}{\beta} \|\nabla G(\theta) - \nabla G(\eta)\|_2^2 \leq (\nabla G(\theta) - \nabla G(\eta))^\top (\theta - \eta).$$

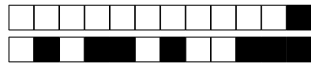
(a) Suppose first that F is quadratic, so that $\nabla F(\theta) = H(\theta - \eta_*)$ with constant Hessian $H = \nabla^2 F$, a symmetric positive-definite matrix whose eigenvalues lie in $[\mu, L]$. Show that

$$\cos \alpha \geq \frac{2\sqrt{\mu L}}{L + \mu}.$$

(b) Show that the same bound holds for an arbitrary (not necessarily quadratic) μ -strongly-convex, L -smooth function F . *Hint: apply co-coercivity to $G(\theta) = F(\theta) - \frac{\mu}{2}\|\theta\|_2^2$, which is convex and $(L - \mu)$ -smooth.*

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